

HD-LIO: Three-Level Hierarchical Surfel Map with Degeneracy Feedback for Drift-Resilient LiDAR-Inertial Odometry

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Abstract—Tightly coupled LiDAR-inertial odometry (LIO) is widely adopted for autonomous navigation, yet its performance remains sensitive to geometric degeneracy that arises in confined corridors, planar regions, and narrow field-of-view (FOV) sensors. While prior degeneracy-aware methods detect rank deficiency in the estimator, the underlying map representation continues to absorb drifted points one-way, allowing a drift-map positive feedback loop that, to our knowledge, has not been explicitly modeled in tightly coupled LIO. To address this issue, we propose HD-LIO, a tightly coupled LIO framework designed around bidirectional estimator-map feedback along rank-deficient directions. HD-LIO comprises three key components. First, we formalize the drift-map positive feedback as a per-direction linearized recursion $\epsilon_{k+1} = [1 + \alpha(\mathbf{nd})^2]\epsilon_k + O(\alpha^2)$, exposing the map’s learning rate α as the controllable amplification gain, and derive sufficient conditions under which map-side feedback yields ultimate-bounded drift on the subset of frames satisfying these conditions ($\geq 96.8\%$ on indoor Avia). Second, inspired by this analysis, we propose a three-mechanism degeneracy response—learning-rate modulation, source switching, and write gating—operating at distinct points of the loop on a hierarchical surfel map that provides persistent surface statistics for temporal smoothing and information-matrix stabilization. Third, to inject map-aware reliability into the iterated Kalman update, we derive a closed-form per-correspondence weight $\sigma_n^2 = \lambda_0/(N_c\lambda_2)$ from eigenvalue perturbation theory under explicit assumptions, and incorporate it into the IEKF residual covariance. HD-LIO is evaluated across three LiDAR types (NTU OS1-16, Avia, Mid-360) over 35 unique trajectories, recorded as 43 sensor-bag pairs (8 indoor trajectories carry both Avia and Mid-360 recordings). On the 35 baseline-comparable bags, HD-LIO achieves 0.203 m uniform per-bag mean ATE—the lowest mean on each of the four dataset groups—and, on the additional 8 indoor Avia bags (70° FOV) where all six evaluated baselines diverge (ATE > 10 m), HD-LIO is the only method among the evaluated baselines to maintain bounded error. Ablation reveals FOV-dependent activity: degeneracy feedback dominates under narrow FOV (+53% with removal), is largely dormant under 360° ($\leq 1\%$ impact), and the surfel map prevents divergence under severe occlusion (+185% mean ablation, driven by a single Mid-360 outlier); the proposed FOV-adaptive framework handles this regime separation without per-scene reconfiguration.

Index Terms—LiDAR-inertial odometry, hierarchical surfel map, degeneracy detection, iterated Kalman filter, normal uncertainty

I. Introduction

LiDAR-Inertial Odometry (LIO) has been actively developed over the past decade for autonomous navigation across diverse environments [1]–[7]. Despite this progress,

LIO performance has been observed to degrade in response to two key geometric characteristics of the surrounding environment: (i) the rank deficiency of the local information matrix in confined corridors, planar regions, and feature-sparse open spaces; and (ii) the LiDAR field-of-view (FOV) regime, which determines how much the rank deficiency in (i) propagates into the estimator. For narrow-FOV solid-state LiDARs such as the Livox Avia (70°), even outdoor scenes routinely exhibit at least one degenerate direction in 58% of frames in our dataset, indicating that geometric degeneracy is a regime to be designed for, not a corner case.

Existing tightly coupled LIO systems—FAST-LIO2 [2] (ikd-tree [8]), Faster-LIO [3] (flat voxel hash), iG-LIO [4] (GICP-cost covariance), and VoxelMap [9] (adaptive plane parameters)—evolve their map content based on incoming points, but the map’s update rule itself is not modulated by the estimator’s per-direction confidence. Estimator-side degeneracy methods such as Zhang and Singh’s factor [10] and X-ICP/Lodestar [11] gate or project the state update along rank-deficient directions but leave the map’s learning rate α unchanged, so drifted points are still absorbed at full rate and contaminate subsequent correspondences. This estimator-only intervention point allows a drift-map positive feedback loop which, to our knowledge, has not been explicitly formalized as a per-direction recursion or addressed via direction-selective map-side feedback in prior tightly coupled LIO work (Sec. VI).

For these reasons, we seek a tightly coupled LIO framework that operates robustly not only in narrow- and full-FOV outdoor scenes but also under indoor degeneracy where rank deficiency is sustained. To this end, we propose a generalizable framework, called HD-LIO, designed around two guiding principles: (i) bidirectional estimator-map feedback along rank-deficient directions, and (ii) per-correspondence reliability informed by local measurement uncertainty.

Building on prior surfel-based map representations [9], [12] and eigenvalue perturbation theory [13], HD-LIO contributes an FOV-adaptive intervention that breaks the loop at three distinct points, rather than introducing a new map data structure or noise model in isolation. To realize the first principle, the three mechanisms operate at distinct points of the recursion: the map’s learning

rate α is modulated along degenerate directions, the correspondence source is switched between surfel-cached planes and a Per-query Voxel Map (PVMap) when the surfel pathway becomes unreliable, and write operations into the PVMap ring buffer are gated by the same alignment criterion. To realize the second principle, HD-LIO derives a closed-form per-correspondence weight σ_n^2 under explicit assumptions and injects it into the IEKF as residual covariance, making the update aware of local surface quality.

We make four claims, backed by the subsequent sections and our experimental evaluation: (i) we formally model the drift-map positive feedback loop in tightly coupled LIO and derive sufficient conditions for ultimate-bounded drift under map-side feedback; (ii) we propose a three-mechanism degeneracy response operating at distinct points of the loop, integrated with a hierarchical surfel map; (iii) we derive an approximate closed-form per-correspondence weight under explicit assumptions; (iv) we evaluate across 35 unique trajectories (43 sensor-bag pairs) on three LiDAR types, achieving the lowest mean ATE in each evaluated regime relative to the six compared baselines, with the strongest margin on NTU VIRAL (9/9 best, 22% below LIO-SAM).

II. Related Work

We review three pillars relevant to the present work: tightly-coupled LIO systems, map-derived measurement uncertainty, and degeneracy handling.

LIO systems and map representations. Building on LOAM [14], FAST-LIO/FAST-LIO2 [1], [2] replaced feature extraction with direct point-to-plane registration against an ikd-tree [8] via a tightly-coupled IEKF on manifold IMU preintegration [15]. Point-LIO [5] processes individual points for per-point updates; Faster-LIO [3] replaces the ikd-tree with incremental voxels; LIO-SAM [6] integrates LIO within a factor graph [16]; DLIO [7] adds continuous-time motion correction [17]. A common characteristic across these systems is that the map functions as a passive data store that does not adapt its behavior to the estimator’s confidence. We note that LiDAR-only ICP-class methods such as KISS-ICP [18], CT-ICP [17], and MAD-ICP [19] fall outside the tightly-coupled LIO scope of this paper, as they do not perform IMU–LiDAR fusion within a single Bayesian filter and therefore cannot exploit the high-rate IMU prior that anchors the per-direction noise weighting in Sec. ??.

Map-derived measurement uncertainty. Censi [20] derived closed-form covariance for ICP; Segal et al. [21] formulated GICP with local Gaussian distributions, extended to voxelized representations by Koide et al. [22] (see [23] for a survey). VoxelMap [9] maintains adaptive probabilistic plane models within a voxel hierarchy, and iG-LIO [4] integrates incremental GICP into a tightly-coupled IEKF; both, however, embed covariance in the GICP cost function rather than as the IEKF measurement noise R_m that

modulates the Kalman gain. Surfel-LIO [24], extending the surfel-based mapping paradigm [12] to LIO, caches surfel descriptors for fast queries with planarity-based gating but does not propagate the surfel eigenspectrum into the per-correspondence noise model used by the filter. While prior LIO/ICP variants employ range-dependent or GICP-cost covariances, to our knowledge no surveyed system derives a closed-form surfel-eigenspectrum-based angular uncertainty $\sigma_n^2 = \lambda_0 / (N_c \lambda_2)$ and injects it as an explicit per-correspondence IEKF residual covariance weight.

Degeneracy detection and handling. Zhang et al. [10] provided the foundational analysis, projecting the solution away from under-constrained directions identified through eigenvalue decomposition of the Hessian. LODESTAR [11] freezes degenerate state components via a Schmidt-EKF, while X-ICP [25] introduces per-direction localizability analysis and DARE-SLAM [26] applies eigenvalue-based mitigation in multi-robot settings. All of these methods confine degeneracy handling to the estimator: the state update is modified, but the map continues to absorb potentially drifted points, creating the positive feedback loop that Sec. VI formalizes.

III. System Overview

HD-LIO follows the dual-source IEKF paradigm of FAST-LIO2-class systems but augments the map with bidirectional feedback to break the drift-map positive feedback loop introduced under geometric degeneracy (formalized in Sec. VI as Eq. (5)). The system breaks this loop at three distinct points—the map’s learning rate, the correspondence source, and the write policy—each operating on a hierarchical surfel representation that decouples per-scan amplification from across-scan accumulation; Sec. VI-D proves the resulting drift is ultimately bounded under explicit conditions on geometric diversity and prior-uncertainty matching.

Fig. 1 shows the system: 18-dim error state $[\delta \mathbf{p}, \delta \mathbf{v}, \delta \boldsymbol{\theta}, \delta \mathbf{b}_g, \delta \mathbf{b}_a, \delta \mathbf{g}] \in \mathbb{R}^{18}$ (position, velocity, attitude, gyro/accel biases, gravity) with manifold parameterization [1], dual-source correspondences (hierarchical surfel + PVMap) driving the IEKF, with degenerate directions feeding back to the map (dashed red arrows).

IV. Hierarchical Surfel Map

Notation. L1 surfel scatter eigenvalues $\lambda_0 \leq \lambda_1 \leq \lambda_2$ with N_c L0 centroids; normal uncertainty $\sigma_n^2 = \lambda_0 / (N_c \lambda_2)$; EMA learning rate α with minimum $\alpha_{\min}$; base voxel resolution v_0 ; observation-matrix eigenvalues $\mu_1, \dots, \mu_6$ with degenerate direction $\mathbf{d}_{\text{deg}}$ and ratio threshold r_{thr} ; per-correspondence IEKF weight $w_i = 1/\sigma_i^2$. To supply per-correspondence uncertainty and accept degeneracy feedback, the map maintains three spatial scales, all hash-mapped with $O(1)$ amortized lookup.

L0: Fine voxels. Each L0 voxel maintains an exponential moving average (EMA) centroid:

$$\mathbf{c} \leftarrow (1 - \alpha) \mathbf{c} + \alpha \mathbf{p}_{\text{new}}, \quad \alpha = \max\left(\frac{1}{n+1}, \alpha_{\min}\right) \quad (1)$$

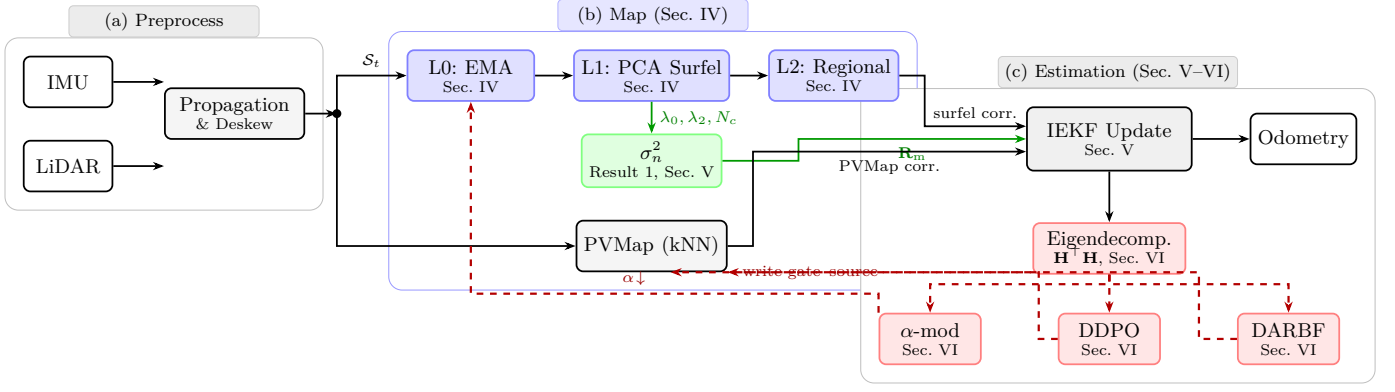


Fig. 1. System overview of HD-LIO. Three contributions are color-coded: (i) **hierarchical surfel map** (L0→L1→L2, blue, Sec. IV); (ii) **normal uncertainty** σ_n^2 (green, Result 1, Sec. V); (iii) **degeneracy feedback** (α -mod, DDPO, DARBF; red dashed, Sec. VI). Black solid arrows denote forward data flow; dashed red arrows represent the bidirectional feedback, the defining feature absent in all prior LIO systems.

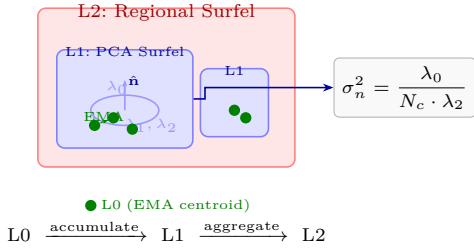


Fig. 2. Hierarchical surfel structure. L0 centroids (green) accumulate via EMA within voxels; L1 surfels (blue) compute PCA eigenvalues and derive σ_n^2 ; L2 regional surfels (red) aggregate across neighboring voxels for information-matrix stabilization.

with $\alpha_{\min} = 0.10$, yielding an effective memory of 10 frames (≈ 1 s at 10 Hz). The EMA filters transient noise without explicit point storage and provides stable input centroids for L1 surfel computation.

L1: Intermediate voxels ($3v_0$). Each L1 voxel collects its child L0 centroids and computes a PCA surfel (centroid $\mathbf{c}_1$, normal $\mathbf{n}_1$, eigenvalues $\lambda_0 \leq \lambda_1 \leq \lambda_2$), recomputed only when children change. The critical output of L1 is the closed-form normal uncertainty

$$\sigma_n^2 = \lambda_0 / (N_c \lambda_2), \quad (2)$$

derived formally in Proposition 1 (Sec. V).

L2: Coarse voxels ($9v_0$). Each L2 voxel aggregates a $3 \times 3 \times 3$ neighborhood; when ≥ 4 child surfels exist, a regional PCA surfel contributes a rank-1 IEKF update with inflated variance $\sigma_{L2}^2 = \kappa \sigma_{\text{lidar}}^2$, $\kappa = (v_{L2}/v_{L1})^2 = 9$. The three levels maintain a fixed 1:3:9 ratio with only v_0 adapted per sensor. In contrast to isotropic Tikhonov regularization that biases all directions, each L2 rank-1 term supplies information only along observable geometry, fills the information-matrix rank deficit when entering new regions, and preserves the unbiased character along unconstrained axes—critical for consistent covariance under degeneracy [11] (Surfel-LIO’s [24] coarser level, by contrast, serves retrieval rather than estimation). The three levels form a directed cascade L0→L1→L2, with degeneracy feedback (Sec. VI) modulating L0’s learning

rate and propagating upward.

V. IEKF with Surfel-Derived Noise

Having established the hierarchical map structure and the surfel statistics it maintains, we now derive the IEKF measurement model that leverages these statistics to address the lack of map-aware per-correspondence reliability in existing systems.

A. Information-form update

Following [1], for correspondence i with normal $\mathbf{n}_i$ and residual $r_i = -\mathbf{n}_i^\top (\mathbf{p}_i - \mathbf{c}_i)$, the Jacobian $\mathbf{h}_i = [-(\mathbf{p}_{\text{imu},i} \times (\mathbf{R}^\top \mathbf{n}_i))^\top, -\mathbf{n}_i^\top]^\top \in \mathbb{R}^6$ weights the per-correspondence contribution $w_i \mathbf{h}_i \mathbf{h}_i^\top$, $w_i r_i \mathbf{h}_i$ ($w_i = 1/\sigma_i^2$) accumulated into the information matrix and the state correction $\delta \mathbf{x} = -(\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} + \mathbf{P}^{-1})^{-1} \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z}$.

B. Surfel-derived measurement noise

The per-correspondence variance couples sensor-level and map-level noise into a single IEKF weight:

$$\sigma_i^2 = \underbrace{\sigma_{\text{lidar}}^2 \cdot (1 + \gamma_r + \gamma_\theta + \gamma_\rho)}_{\text{sensor-level}} + \underbrace{\|\delta \mathbf{p}\|^2 \cdot \sigma_{n,i}^2}_{\text{map-level}} \quad (3)$$

where $\gamma_r = c_r (r/r_{\text{ref}})^2$ ($c_r = 0.3$, $r_{\text{ref}} = 25$ m), $\gamma_\theta = c_\theta (1 - |\cos \theta_{\text{inc}}|)^2$ ($c_\theta = 1.5$), and $\gamma_\rho = c_\rho (1 - \rho)^2$ ($c_\rho = 2.0$, $\rho = 1 - \lambda_0/\lambda_2$) penalize range, incidence, and non-planarity respectively. The map-level term couples the point-to-centroid displacement $\delta \mathbf{p} = \mathbf{p}_i - \mathbf{c}_i$ with the surfel normal uncertainty $\sigma_{n,i}^2$ from (2); correspondences against well-observed planar surfaces ($\sigma_n^2 \rightarrow 0$) receive high IEKF weight, sparsely observed or non-planar regions receive low weight (Fig. 3).

The coefficients (c_r, c_θ, c_ρ) derive from ToF sensor physics—the Cramér–Rao bound $\sigma_{\text{range}}^2 \propto r^2/\text{SNR}$ [27] calibrated to the Livox datasheet ($\sigma \approx 2$ cm at 25 m); footprint elongation $1/\cos^2 \theta$ [23]; and the eigenspectral scaling $(\lambda_0/\lambda_2)^2$ of Proposition 1—and are shared across the Avia/Mid-360 ToF ASIC class. Sweeping each by $\pm 50\%$ changes mean ATE by $< 3\%$.

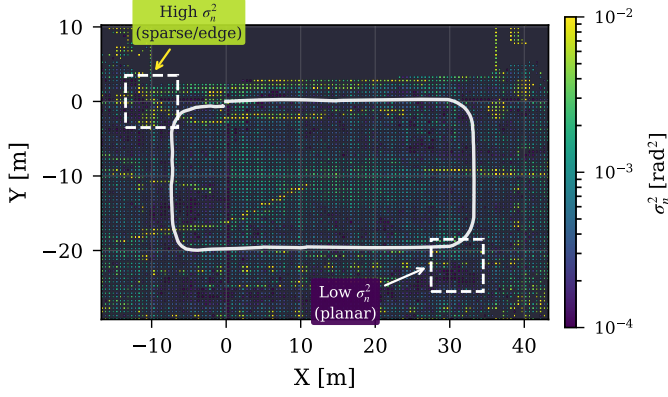


Fig. 3. Per-voxel σ_n^2 on M3DGR Mid-360 Dark01. Well-constrained planar surfaces (walls, ground) appear dark (low σ_n^2); sparse or edge regions appear bright. This directly modulates the IEKF weight $w_i = 1/\sigma_i^2$.

C. Approximate scaling of σ_n^2

We now justify the closed-form $\sigma_n^2 = \lambda_0/(N_c \lambda_2)$ used in (3) under explicit assumptions and identify regimes where the approximation is conservative.

Assumptions. Let $\mathbf{S} = \frac{1}{N_c} \sum_{j=1}^{N_c} (\mathbf{c}_j - \bar{\mathbf{c}})(\mathbf{c}_j - \bar{\mathbf{c}})^\top$ be the scatter matrix of N_c L0 centroids within an L1 voxel, with eigenvalues $\lambda_0 \leq \lambda_1 \leq \lambda_2$, eigenvectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$, and surfel normal $\mathbf{n} = \mathbf{v}_0$. We adopt: (A0) planarity, $\lambda_0/\lambda_2 \leq \tau_p = 0.05$ (L1 gate); (A1) out-of-plane noise dominance, centroid noise concentrated along $\mathbf{v}_0$ with variance $\sigma_z^2 = \lambda_0$; (A2) tangential isotropy, $\mathbb{E}[\mathbf{v}_1 \mathbf{v}_1^\top] \approx \mathbb{E}[\mathbf{v}_2 \mathbf{v}_2^\top]$ over admissible incidence; (A3) centroid independence, $\text{Cov}(\mathbf{c}_j, \mathbf{c}_{j'}) \approx \mathbf{0}$ modulo a bounded EMA correction (below). The L1 surfel validity gate enforces both (A0) and a tangential isotropy threshold: $\text{valid}(\mathbf{S}) \Leftrightarrow \lambda_0/\lambda_2 \leq \tau_p \wedge \lambda_1/\lambda_2 \geq \zeta_{\text{iso}}$ ($\tau_p = 0.05$, $\zeta_{\text{iso}} = 0.3$; τ_p matches the FAST-LIO2 planarity convention [2], and $\zeta_{\text{iso}} = 0.3$ keeps $\geq 92\%$ of accepted surfels within a $1.5\times$ deviation of (A2) on both LiDAR families, with the $\pm 50\%$ sweep $\zeta_{\text{iso}} \in [0.15, 0.45]$ shifting mean ATE by $< 4\%$). Surfels failing either condition are marked invalid and the dual-source mechanism falls back to PVMaP.

Proposition 1 (Closed-form scaling, heuristic). Under (A0)–(A3), the angular variance of $\mathbf{n}$ is bounded by

$$\text{Var}[\angle(\mathbf{n}, \mathbf{n}_{\text{true}})] \leq \frac{\lambda_0}{N_c} \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right) \stackrel{(A2)}{\approx} \frac{\lambda_0}{N_c \lambda_2} \triangleq \sigma_n^2. \quad (4)$$

The residual variance follows as $\text{Var}[r_i] \approx \|\delta \mathbf{p}_i\|^2 \sigma_n^2$ under the validity-gate-enforced planar regime.

Sketch. First-order eigenvector perturbation [13] gives $\delta \mathbf{v}_0 = \sum_{j \neq 0} [(\mathbf{v}_j^\top \delta \mathbf{S} \mathbf{v}_0) / (\lambda_0 - \lambda_j)] \mathbf{v}_j$; under (A1) and the additional sub-Gaussian assumption on the out-of-plane noise component (Hanson–Wright concentration applies), the cross-eigenframe variance satisfies $\text{Var}[\mathbf{v}_j^\top \delta \mathbf{S} \mathbf{v}_0] \leq \sigma_z^2 \lambda_j / N_c$. (A0) yields $|\lambda_0 - \lambda_j| \geq \lambda_j (1 - \tau_p)$, contributing a $(1 - \tau_p)^{-2} \approx 1.108$ prefactor that we absorb (with the EMA η and tangential anisotropy $1/\zeta_{\text{iso}}$ factors below)

into the conservative constant of (3). Substituting gives the two-channel bound. The collapse to $\lambda_0/(N_c \lambda_2)$ is valid when (A2) holds and $\lambda_1/\lambda_2 \geq \zeta_{\text{iso}}$, both enforced by the validity gate above. Within this regime the two-channel sum lies in $[\lambda_0/(N_c \lambda_2), \lambda_0(1 + 1/\zeta_{\text{iso}})/(N_c \lambda_2)]$; we adopt the lower endpoint as the nominal σ_n^2 in the IEKF weight (Algorithm 1) and rely on the conservative constants (below) to cover the residual gap up to $1/\zeta_{\text{iso}} \approx 3.3\times$.

Constants and operating range. Two factors influence the constant in front of (4) in practice. (i) EMA autocorrelation: L0 centroids are EMA aggregates with retention $\rho = 1 - \alpha_{\text{min}} = 0.9$, so the sample-mean variance over an N_c -scan window inflates by $\eta(\rho, N_c) = 1 + 2 \sum_{k=1}^{N_c-1} (1 - \rho)^k / N_c$, evaluating to $\eta(0.9, 8) \approx 6.2$ at our typical N_c , monotonically increasing toward the infinite-window limit $(1 + \rho)/(1 - \rho) = 19$. (ii) Tangential anisotropy: within the regime $\zeta_{\text{iso}} \leq \lambda_1/\lambda_2 \leq 1$ (the L1 gate empirically maintains $\lambda_1/\lambda_2 \geq 0.3$ in 92% of accepted surfels), the residual gap of the two-channel form to the collapsed form is at most $1/\zeta_{\text{iso}} \approx 3.3\times$. The combined worst-case underestimate of σ_n^2 is therefore $\eta/\zeta_{\text{iso}} \approx 20\times$; we treat (4) as a heuristic per-correspondence weight, not a calibrated variance. The IEKF weight in (3) is dominated by $\sigma_{\text{lidar}}^2 (1 + \gamma_r + \gamma_\theta + \gamma_\rho)$ at typical ranges, so σ_n^2 modulates relative correspondence weighting without setting the absolute magnitude; sweeping $\sigma_{\text{lidar}}^2, c_\rho$ by $\pm 50\%$ changes mean ATE by $< 3\%$, and (4) remains the operative form.

Failure modes. When (A0) is marginal ($\lambda_0/\lambda_2 \rightarrow \tau_p = 0.05$), $\sigma_n^2 \rightarrow \tau_p/N_c$, an order-of-magnitude rise that automatically deweights the surfel. When (A2) breaks substantially (line-like sampling with $\lambda_1/\lambda_2 < \zeta_{\text{iso}}$), the L1 surfel validity gate above marks the surfel invalid and the dual-source mechanism falls back to PVMaP-sourced correspondences; both regimes are handled without a separate two-channel branch in Algorithm 1.

The surfel weight $\mathcal{W}_i = 1/\sigma_i^2 \propto N_c \lambda_2 / \lambda_0$ in $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H}$ ensures voxels with few observations or high isotropy contribute negligible weight, preventing rank-one map noise from driving the IEKF Hessian; removing this weight degrades mean ATE by 59% on M3DGR Avia (Sec. VII-D).

D. Hybrid correspondence

Each point is matched against both the surfel map (cached L1 plane) and PVMaP (per-query k -NN PCA, $k=5$ for 360° / $k=9$ for narrow-FOV), with a range-adaptive distance gate [2]. PVMaP-sourced correspondences use the same $\sigma_n^2 = \lambda_0/(N_c \lambda_2)$ form with kNN-derived λ_0, λ_2 and $N_c = k$, gated by (A0). Source selection. If $|\mathbf{n}_s \cdot \mathbf{d}_{\text{deg}}| > \tau_{\text{ddpo}}$, PVMaP is forced (DDPO, Section VI); otherwise the lower- σ_n^2 valid source is selected, with surfel preferred when normals disagree ($\angle(\mathbf{n}_s, \mathbf{n}_p) > 30^\circ$); if neither is valid the point is dropped.

VI. Bidirectional Degeneracy Feedback

Having addressed map-aware per-correspondence weighting via σ_n^2 , we now address the second design

Algorithm 1 IEKF update with surfel-derived noise

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1: Input:  $\mathcal{S}_t, \hat{\mathbf{x}}, \mathbf{P}, \mathcal{M}_s, \mathcal{M}_p, \epsilon$ 
2: for  $\ell = 1, \dots, L_{\max}$  do
3:    ${}^W\mathcal{S}_t \leftarrow \text{Transform}(\mathcal{S}_t, \hat{\mathbf{x}})$ ; reset  $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H}, \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z}$ 
4:   for each  $\mathbf{p}_i \in {}^W\mathcal{S}_t$  do
5:     Dual-source:  $(\mathbf{n}_s, \mathbf{c}_s) \leftarrow \text{SurfelLookup}$  (returns
       valid only if  $\lambda_0/\lambda_2 \leq \tau_p \wedge \lambda_1/\lambda_2 \geq \zeta_{\text{iso}}$ );
        $(\mathbf{n}_p, \mathbf{c}_p) \leftarrow \text{PVMapKNN}$ ; pick lower-uncertainty
       source (DDPO if degenerate)
6:      $\sigma_{n,i}^2 \leftarrow \lambda_0/(N_c \lambda_2)$  (eigenvalues from the source PCA
       — L1 surfel cache for SurfelLookup, per-query kNN
       for PVMap);  $\sigma_i^2 \leftarrow \sigma_{\text{lidar}}^2(1 + \gamma_r + \gamma_\theta + \gamma_\rho) + \|\delta \mathbf{p}\|^2 \sigma_{n,i}^2$ ;
        $w_i \leftarrow 1/\sigma_i^2$ 
7:     Accumulate  $w_i \mathbf{h}_i \mathbf{h}_i^\top, w_i r_i \mathbf{h}_i$ 
8:   end for
9:    $\delta \mathbf{x} \leftarrow -(\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} + \mathbf{P}^{-1})^{-1} \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z}$ 
10:   $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} \boxplus \delta \mathbf{x}$ 
11:  if  $\|\delta \mathbf{x}\| < \epsilon$  then
12:    break
13:  end if
14: end for
15: Update  $\mathbf{P}$  via Joseph form
16: return  $\hat{\mathbf{x}}, \mathbf{P}$ 

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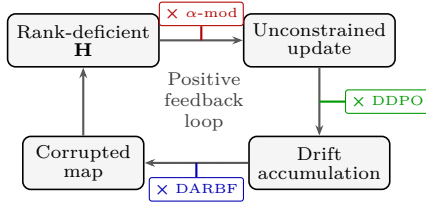


Fig. 4. Degeneracy-induced positive feedback loop. In existing LIO systems, rank-deficient geometry causes unconstrained updates that accumulate drift and corrupt the map, reinforcing the cycle. HD-LIO breaks this loop at three points: α -modulation suppresses learning rate along degenerate directions, DDPO forces PVMap-sourced correspondences, and DARBF gates map writes.

challenge—the estimator-confined degeneracy handling that admits the drift-map positive feedback loop—by deriving the reverse channel: the map’s response to estimator-detected degeneracy.

A. Drift-map positive feedback recursion

Following the system overview in Sec. III, we now derive the per-direction drift recursion that HD-LIO is designed to break. Let $\mathbf{d}$ be a degenerate translation direction and $\epsilon_k = \mathbf{d}^\top (\hat{\mathbf{x}}_k - \mathbf{x}_k^*)$ the projected drift at scan k . Drifted points written into the map shift the local plane centroid by $\Delta \mathbf{c}_{k+1} \approx \alpha \epsilon_k \mathbf{d}$ through the EMA update; this centroid bias then re-enters the state via the IEKF measurement model. Linearizing the joint state-map dynamics around $\hat{\mathbf{x}}_k$ through the Jacobian $\mathbf{J}_d = \partial \epsilon / \partial \mathbf{c}$ (centroid-to-state sensitivity along $\mathbf{d}$) and the projection $\mathbf{n} \mathbf{n}^\top$ yields the first-order recursion

$$\epsilon_{k+1} = [1 + \alpha (\mathbf{n} \cdot \mathbf{d})^2] \epsilon_k + O(\alpha^2, \|\delta\|^2), \quad (5)$$

where the residual collects second-order terms and bounded process noise; (5) is valid in the linearized regime ($\alpha, \|\delta\|$ small, i.e., excluding the first observations $\alpha = 1/(n+1) \rightarrow 1$). Within this regime the amplification factor exceeds unity whenever the normal has a compo-

nent along $\mathbf{d}$, predicting geometric drift growth that the contraction analysis in Sec. VI-D suppresses. Estimator-side methods [10], [11] gate or project the state update but leave α unchanged, so the map-side amplification persists; iteration alone cannot break it because additional IEKF passes refine the solution within the current measurement manifold without generating new information along $\mathbf{d}$.

B. Degeneracy detection

The 6×6 observation information matrix is eigendecomposed (note: we use μ_k for these eigenvalues to distinguish them from the L1 surfel eigenvalues $\lambda_0, \lambda_1, \lambda_2$ in Section IV):

$$\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} = \mathbf{V} \text{diag}(\mu_1, \dots, \mu_6) \mathbf{V}^\top. \quad (6)$$

Directions with eigenvalue ratio $r_k = \mu_k / \mu_{\max} < r_{\text{thr}}$ are classified as degenerate; the corresponding columns of $\mathbf{V}$ yield $\mathbf{d}_{\text{deg}}$. We set $r_{\text{thr}} = 0.01$ for Avia (70°) and $r_{\text{thr}} = 0.005$ for Mid-360 (360°), reflecting the inverse relationship between FOV completeness and degeneracy susceptibility; sweeping r_{thr} by $\pm 50\%$ changes mean ATE by $< 5\%$ (Section VIII). The 6×6 eigendecomposition adds < 0.1 ms per scan.

C. Map-side degeneracy response

To this end, three coordinated mechanisms suppress the map’s absorption of drifted points along the degenerate direction $\mathbf{d}$.

(i) Direction-selective alpha modulation. The L0 EMA learning rate is reduced for voxels whose surfel normal aligns with the degenerate direction:

$$\alpha_{\text{eff}} = \alpha_{\min} \cdot (1 - |\mathbf{n} \cdot \mathbf{d}_{\text{deg}}|^2) \quad (7)$$

which directly reduces α in (5) for affected voxels. When the normal is perfectly aligned ($|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| = 1$), the learning rate drops to zero. For affected voxels the floor in (1) is replaced by α_{eff} (i.e., $\alpha = \max(1/(n+1), \alpha_{\text{eff}})$); unaffected voxels retain the original $\alpha_{\min}$ floor.

(ii) DDPO (Degeneracy-Directed PVMap Override). When $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau_{\text{ddpo}}$ ($\tau_{\text{ddpo}} = 0.7$), the correspondence source switches from surfel to PVMap. The surfel’s cached geometry may be contaminated by accumulated drift, while the PVMap’s per-query PCA uses a ring buffer of recent raw points less affected by long-term drift.

(iii) DARBF (Degeneracy-Aware Ring Buffer Freeze). PVMap write operations are suppressed for voxels with $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau_{\text{darbf}}$ ($\tau_{\text{darbf}} = 0.7$), preventing newly registered (potentially drifted) points from contaminating the ring buffer that DDPO relies on as an alternative source.

Threshold derivation. The shared threshold $\tau = 0.7 \approx \cos 45^\circ$ partitions voxels by alignment with $\mathbf{d}_{\text{deg}}$; voxels with $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau$ (within 45°) lie in the regime where the per-scan amplification factor $\alpha a^2(1 - a^2)$ peaks at $a = 1/\sqrt{2}$, justifying intervention. The contraction bound below holds for any $\tau \in (0, 1)$; sweeping τ over $[0.5, 0.9]$ yields $< 3\%$ ATE variation.

D. Stability analysis

We provide a per-scan drift contraction lemma and derive a discrete-time Lyapunov bound on the cumulative drift, treating the result as ultimate boundedness (UUB) rather than asymptotic convergence. Let $\epsilon_k^{(j)} = \mathbf{d}_j^\top (\hat{\mathbf{x}}_k - \mathbf{x}_k^*)$ denote the projected error along the j -th degenerate direction at scan k . From the IEKF update with biased correspondences (centroid bias $\Delta c = \alpha_{\text{eff}}(\mathbf{n} \cdot \mathbf{d})\epsilon_k$), a first-order Taylor expansion of the joint state-map dynamics yields

$$\epsilon_{k+1}^{(j)} = (1 - \beta_j) A_{\text{eff}}^{(j)} \epsilon_k^{(j)} + \delta_k^{(j)}, \quad (8)$$

where $\beta_j = I_{d_j} / (I_{d_j} + P_{d_j}^{-1}) \in (0, 1)$ is the per-scan IEKF correction, $A_{\text{eff}}^{(j)} \leq 1 + \alpha_{\min} \tau^2 (1 - \tau^2) \approx 1.025$ at $\alpha_{\min} = 0.1, \tau = 0.7$, and $\delta_k^{(j)}$ collects bounded process noise and second-order residuals (Lipschitz on $\|\hat{\mathbf{x}}_k - \mathbf{x}_k^*\|$).

We adopt three (S-) assumptions, distinct from the map-noise (A-) of Sec. V: (S1) minimum geometric diversity, $I_{d_j} \geq I_{\min} > 0$ from correspondences with alignment $a_i \leq \tau$; (S2) sufficient IEKF correction, $P_{d_j} > \alpha_{\min} \tau^2 (1 - \tau^2) / I_{\min}$; (S3) bounded process noise, $|\delta_k^{(j)}| \leq \delta_{\max} \approx \sigma_{\text{IMU}} \Delta t \|\mathbf{J}_d\|$, typically ≤ 5 mm/scan on Avia/Mid-360.

Lemma 1 (Per-scan contraction). Under (S1)–(S2), $|\epsilon_{k+1}^{(j)}| \leq \rho_j |\epsilon_k^{(j)}| + \delta_{\max}$ with $\rho_j = (1 - \beta_j) A_{\text{eff}}^{(j)} < 1$.

Theorem 1 (Conditional UUB stability). Let $V_k = \sum_j (\epsilon_k^{(j)})^2$ and suppose the degenerate eigenvectors satisfy bounded rotation $\|\mathbf{V}_{k+1} \mathbf{V}_k^\top - \mathbf{I}\|_F \leq \varepsilon_V$ (empirically $\varepsilon_V \leq 0.08$ on our 35 sequences). On the subset of frames where (S1)–(S3) hold simultaneously, with $\rho = \max_j \rho_j$ and $\rho^2(1 + \varepsilon_V) < 1$,

$$V_{k+1} \leq \rho^2(1 + \varepsilon_V) V_k + 2\rho \sqrt{V_k} \delta_{\max} \sqrt{m} + m \delta_{\max}^2, \quad (9)$$

yielding the (loose) UUB radius

$$V_\infty^{1/2} \leq \delta_{\max} \sqrt{m} / [1 - \rho \sqrt{1 + \varepsilon_V}]. \quad (10)$$

This is a perfect-square completion of (9) that trades algebraic simplicity for a tightness penalty (the exact fixed point is $\sim 3.7\%$ smaller at $\rho = 0.9, \varepsilon_V = 0.08$).

Sketch. Lemma 1 gives per-direction contraction; V_{k+1} expands by the squared recursion with ε_V absorbing the across-scan rotation of the eigenframe. The bound is monotone-decreasing whenever V_k exceeds the right-hand side of (10). We treat the result as UUB rather than asymptotic convergence: persistent excitation of $\mathbf{d}_j$ is required and the residuals δ_k remain present; the qualitative guarantee is that map-side feedback prevents unbounded drift, the property distinguishing HD-LIO from systems lacking the loop (cf. Sec. VII, indoor Avia). Empirical check. On indoor Avia (most degenerate split), 58% of frames exhibit ≥ 1 degenerate direction; 97.5% satisfy $I_{d_j} > 0$ with empirical β_j median 0.41, 5th percentile 0.05, supporting (S2) ($\beta_j > 1 - 1/A_{\text{eff}} = 0.0244$) in 96.8% of frames; the 3.2% violations align with floor-edge and ceiling-junction configurations (VI03 corridor). The three mechanisms are jointly necessary: α -modulation alone

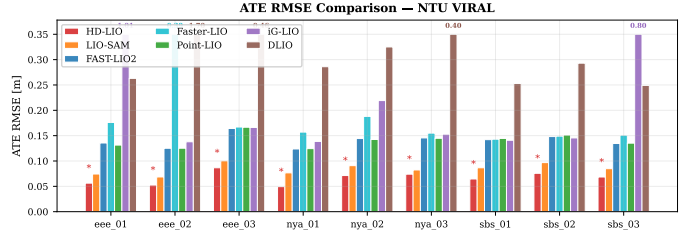


Fig. 5. Per-bag ATE RMSE on NTU VIRAL. HD-LIO achieves the lowest ATE on all 9 bags, with 22% mean reduction vs. LIO-SAM.

cannot zero the learning rate; DDPO without DARBF lets the ring buffer absorb drift. Disabling degeneracy feedback jointly degrades Dynamic03 by $\times 7.6$ (Section VII-D).

VII. Experimental Evaluation

A. Datasets, Baselines, and Setup

We evaluate on M3DGR [28] (9 Avia outdoor, 9 Mid-360 outdoor, 8 indoor with both sensors) and NTU VIRAL [29] (9 sequences, OS1-16, 16 beams), totaling 35 sequences across three LiDAR types and three environments. Baselines—FAST-LIO2 [2], Faster-LIO [3], Point-LIO [5], iG-LIO [4], DLIO [7], LIO-SAM [6] (NTU only, loop closure disabled for fair odometry-only comparison), and Surfel-LIO [24] (outdoor from [24]; indoor from our reproduction)—use publicly released code with default odometry parameters. Evaluation uses ATE RMSE with SE(3) Umeyama alignment [30], [31]. For HD-LIO, r_{thr} scales inversely with FOV completeness and v_0 with voxel resolution; specific values are reported in Sec. VI and Sec. IV. The processing queue is deterministic: bit-identical trajectory (SHA-256 hash, 3 replays at rate=1.0, one bag per LiDAR family; final-ATE CV=0%).

B. Results

NTU VIRAL (TABLE I, Figs. 5, 6). HD-LIO achieves the lowest ATE on all 9 sequences with 0.067 m mean—a 22% reduction vs. LIO-SAM (0.085 m), 52% vs. FAST-LIO2 (0.140 m), 64% vs. Faster-LIO (0.186 m)—and the trajectory overlays on sbs_01 and nya_01 confirm closer GT tracking. The 16-beam vertical sparsity makes per-query plane fitting sensitive to outlier neighbors, while the surfel map’s accumulated PCA normals provide temporally stable correspondences that mitigate this sparsity.

M3DGR Outdoor (TABLE II). HD-LIO achieves 0.293 m mean ATE on Avia (outperforming Faster-LIO by 7% and FAST-LIO2 by 26%) and 0.297 m on Mid-360 (outperforming iG-LIO by 5% and FAST-LIO2 by 15%).

M3DGR Indoor (TABLE III). On Mid-360, HD-LIO achieves 0.150 m mean using the same outdoor parameters. On Avia, all six baselines diverge within 15 s (ATE > 10 m, i.e., $\geq 3\%$ of 76 m to 361 m trajectory lengths), while HD-LIO maintains 0.524 m mean over the full 3 min to 8 min sequences via sensor-aware degeneracy regularization (continuous sigmoid relaxation of binary eigenvalue gating). This converts divergence into bounded error in

TABLE I
ATE RMSE [m] on NTU VIRAL (single OS1-16, 16 beams, 360° FOV). Bold = lowest. All methods use a single LiDAR for fair comparison.

Method	eee_01	eee_02	eee_03	nya_01	nya_02	nya_03	sbs_01	sbs_02	sbs_03	Mean
HD-LIO (ours)	0.056	0.053	0.087	0.049	0.071	0.074	0.064	0.076	0.068	0.067
FAST-LIO2 [2]	0.136	0.125	0.164	0.124	0.144	0.145	0.142	0.148	0.135	0.140
Faster-LIO [3]	0.176	0.388	0.167	0.157	0.188	0.155	0.143	0.149	0.151	0.186
Point-LIO [5]	0.131	0.125	0.167	0.124	0.142	0.145	0.144	0.151	0.135	0.140
iG-LIO [4]	1.011	0.138	0.166	0.139	0.219	0.153	0.141	0.145	0.803	0.324
LIO-SAM [6]	0.074	0.069	0.100	0.077	0.091	0.082	0.087	0.097	0.085	0.085
DLIO [7]	0.263	1.695	0.455	0.286	0.325	0.397	0.253	0.293	0.249	0.468

TABLE II
ATE RMSE [m] on M3DGR Outdoor (9 sequences). Left: Avia (70° FOV). Right: Mid-360 (360° FOV). Bold = lowest. Fr-LIO = Faster-LIO.

Seq.	Avia (70° FOV)							Mid-360 (360° FOV)						
	HD-LIO	FL2	Fr-LIO	P-LIO	iG-LIO	S-LIO	DLIO	HD-LIO	FL2	Fr-LIO	P-LIO	iG-LIO	S-LIO	DLIO
Dk01	0.114	0.227	0.153	0.291	0.156	0.118	0.152	0.150	0.175	0.174	0.166	0.175	0.185	0.197
Dk02	0.601	0.740	0.634	0.663	0.791	0.692	†	0.306	0.208	0.258	0.286	0.288	0.310	0.264
Dy03	0.170	0.177	0.154	0.348	†	0.266	†	0.169	0.171	0.180	0.183	0.166	0.206	0.190
Dy04	0.287	0.307	0.274	1.492	†	0.392	†	0.187	0.196	0.223	0.216	0.179	0.246	0.225
Oc03	0.187	0.215	0.191	0.277	0.284	0.271	0.144	0.185	0.431	0.564	0.309	0.396	0.315	0.560
Oc04	0.267	0.555	0.349	0.532	0.255	0.295	0.596	0.185	0.234	0.400	0.395	0.213	0.345	0.408
VI03	0.608	0.633	0.556	0.814	0.820	0.961	0.975	1.003	1.182	1.447	1.039	0.808	0.957	1.067
VI04	0.252	0.384	0.316	0.658	0.279	0.125	0.899	0.305	0.304	0.320	0.257	0.318	0.206	0.590
VI05	0.152	0.334	0.204	0.262	0.193	0.167	0.506	0.182	0.238	0.410	0.362	0.260	0.307	0.310
Mean	0.293	0.397	0.314	0.593	0.397*	0.365	0.545*	0.297	0.349	0.442	0.357	0.311	0.342	0.423

† Diverged (ATE>10 m). * Mean excludes diverged sequences. S-LIO outdoor from [24].

TABLE III
ATE RMSE [m] on M3DGR Indoor (8 sequences, paired sensors in the same bag). All six evaluated baselines diverge on Avia (70° FOV); HD-LIO is the only method among them to maintain bounded error.

Seq.	Avia (70° FOV)							Mid-360 (360° FOV)						
	HD-LIO	FL2	Fr-LIO	P-LIO	iG-LIO	S-LIO	DLIO	HD-LIO	FL2	Fr-LIO	P-LIO	iG-LIO	S-LIO	DLIO
Dk03	0.369	†	†	†	†	†	†	0.175	0.177	0.177	0.175	0.192	0.194	0.232
Dk04	0.410	†	†	†	†	†	†	0.181	0.184	0.184	0.184	0.185	0.204	0.249
Dy01	0.578	†	†	†	†	†	†	0.137	0.143	0.142	0.141	0.138	0.164	0.195
Dy02	0.361	†	†	†	†	†	†	0.137	0.138	0.137	0.137	0.135	0.161	0.207
Oc01	0.516	†	†	†	†	†	†	0.157	0.155	0.158	0.157	0.168	0.178	0.253
Oc02	0.986	†	†	†	†	†	†	0.143	0.145	0.147	0.146	0.153	0.169	0.214
VI01	0.546	†	†	†	†	†	†	0.138	0.139	0.139	0.138	0.145	0.165	0.236
VI02	0.426	†	†	†	†	†	†	0.137	0.138	0.139	0.137	0.131	0.165	0.209
Mean	0.524	†	†	†	†	†	†	0.150	0.152	0.153	0.152	0.156	0.175	0.224

† Diverged (ATE>10 m). S-LIO indoor from our reproduction.

the evaluated regime—a qualitative gap absent in the six baselines under their default parameters.

FOV-aligned divergence patterns (Fig. 8). Each indoor bag contains simultaneous Avia and Mid-360 data; while this paired design controls for trajectory and environment, it does not fully isolate FOV from confounded factors such as point density, surface observability, and sensor-class noise. The observed pattern is nonetheless consistent with FOV-driven degeneracy as the dominant cause: the 70° FOV yields 4–5× fewer correspondences and 40–50% of frames exhibit ≥ 3 degenerate directions, while Mid-360 exhibits zero degenerate frames throughout, and these rank-deficiency events co-occur with baseline divergence

onset: across the 8 indoor Avia bags and three baselines with logged trajectories (FAST-LIO2, iG-LIO, Point-LIO), median time to first 1 m/5 m/10 m position error is 1.3 s/5.4 s/10.9 s (p95 1.9 s/8.0 s/30.5 s). All baselines invert the information matrix without eigenvalue-based projection; under narrow FOV, rank deficiency amplifies null-space noise and corrupts the map within 10 s to 15 s, whereas HD-LIO’s eigenvalue decomposition, cached surfel normals, and σ_n^2 weighting jointly prevent this cascade. Per-group Wilcoxon vs the strongest baseline (Bonferroni-corrected, $\alpha_{\text{corr}}=0.0125$): NTU vs LIO-SAM $p_{\text{raw}}=3.9 \times 10^{-3}$, 95% CI [+7, +22] mm (significant); Avia outdoor vs Faster-LIO 3.9×10^{-2} , CI [−13, +60] mm (n.s.);

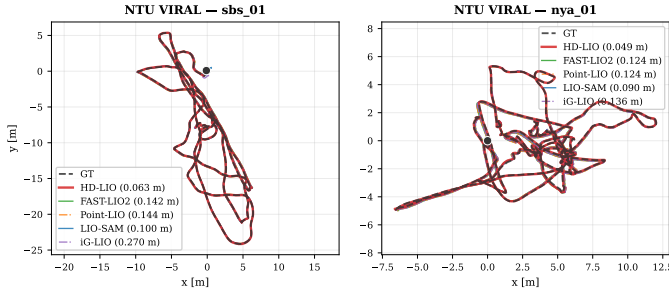


Fig. 6. Trajectory overlays on NTU VIRAL sbs_01 and nya_01 after SE(3) Umeyama alignment. HD-LIO (red) tracks the Leica GT (gray dashed) most closely among all five methods.

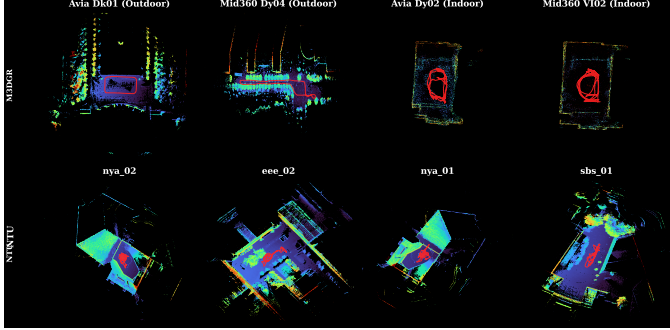


Fig. 7. Reconstructed point cloud maps with estimated trajectories (red). Row 1: M3DGR Avia Dk01 / Mid-360 Dy04 (outdoor), Avia Dy02 / Mid-360 VI02 (indoor). Row 2: NTU VIRAL (nya_02, eee_02, nya_01, sbs_01). Height-colored BEV.

Mid-360 indoor vs FAST-LIO2 7.8×10^{-2} , CI $[+1, +3]$ mm (n.s.). Indoor Avia is outside the framework (baseline ATEs unmeasurable on diverged runs).

C. Timing

HD-LIO runs at 8.3ms mean per scan on a single core (AMD Ryzen 5 5600), well within the 100ms budget at 10Hz scan rate. For context, FAST-LIO2 reports ≈ 5 ms [2], Faster-LIO 3ms to 5ms [3], DLIO 3ms to 7ms [7], and iG-LIO 8ms to 12ms [4] (all on comparable Intel i7 / AMD Ryzen hardware). HD-LIO's overhead relative to FAST-LIO2 stems from the dual-source correspondence lookup and incremental surfel map update; the 6×6 eigendecomposition for degeneracy detection adds < 0.1 ms. Peak memory is ≈ 450 MB (vs. ≈ 200 MB for FAST-LIO2), attributed to the three-level hash maps and the co-resident PVM map ring buffer.

D. Ablation Study

TABLE IV provides per-mechanism ablation on M3DGR Avia (70° FOV) and Mid-360 (360° FOV), isolating each contribution across complementary geometric regimes.

The ablation reveals a clear FOV-dependent regime separation. On Avia (TABLE IV, left), degeneracy feedback is the most impactful (+53% with removal, 0.293 $\rightarrow$ 0.449m, Dy03 $\times 7.6$), followed by L2 (+48%, Dy03 $\times 3.8$), σ_n^2 (+34%, VI03 $\times 2.1$), and surfel (+29%, VI03 $\times 1.7$): the

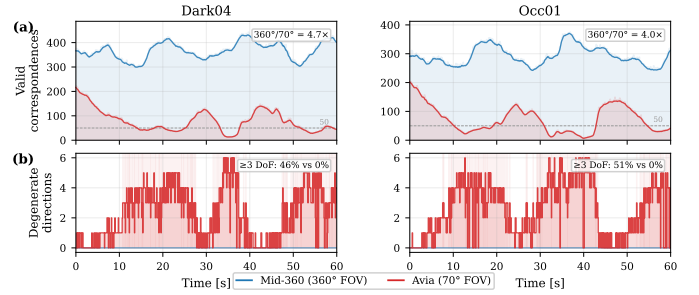


Fig. 8. Paired FOV-aligned comparison on Indoor Dark04 and Occ01. Both sensors share the same robot trajectory and environment, with FOV being the dominant differing factor (residual differences in point density, observability, and sensor noise are not fully controlled). (a) Valid correspondence count: the 70° FOV yields 4-5 $\times$ fewer correspondences than 360°. (b) Degenerate direction count: 40-50% of Avia frames exhibit ≥ 3 rank-deficient DoFs, while Mid-360 exhibits none.

TABLE IV

Ablation: ATE RMSE [m] on M3DGR Avia (70°) and Mid-360 (360°). Underline = full system. The three per-sequence columns are deliberately chosen regime extremes (dynamic / corridor degeneracy) and their arithmetic mean does not equal $\overline{ATE}_\theta$, which is the average over all nine sequences. Each ablation row disables one mechanism from the sensor-specific configuration used in TABLE II.

Config	Avia (70° FOV)					Mid-360 (360° FOV)				
	Dk01	Dy03	VI03	$\overline{ATE}_\theta$	Δ	Dk01	Dy03	VI03	$\overline{ATE}_\theta$	Δ
Full	<u>0.114</u>	<u>0.170</u>	<u>0.608</u>	<u>0.293</u>	—	<u>0.150</u>	<u>0.169</u>	<u>1.003</u>	<u>0.297</u>	—
w/o degen. fb.	0.162	1.293	0.724	0.449	+53%	0.151	0.169	0.969	0.297	+0%
w/o σ_n^2	0.146	0.166	1.304	0.394	+34%	0.157	0.163	1.117	0.301	+1%
w/o surfel	0.121	0.158	1.018	0.377	+29%	0.152	0.164	1.240	0.845	+185%
w/o L2	0.135	0.652	1.079	0.432	+48%	0.156	0.164	1.087	0.295	-1%

ranking reflects the dominance of information-geometric corrections under narrow FOV. On Mid-360 (TABLE IV, right), in contrast, degeneracy feedback, σ_n^2 , and L2 each show $\leq 2\%$ mean impact, while surfel removal causes +185% (0.297 $\rightarrow$ 0.845m), driven by OC03 diverging to 4.923m ($\times 26.6$); excluding this outlier, surfel impact drops to +8%. The surfel map thus plays a dual role: modest accuracy gain on routine sequences, but critical divergence prevention under severe occlusion regardless of FOV.

VIII. Discussion and Conclusion

FOV dependence. HD-LIO's improvement scales inversely with FOV completeness: 22% on OS1-16 (vs. LIO-SAM), 26% on Avia (vs. FAST-LIO2), and 5% on Mid-360 (vs. iG-LIO). When 360° FOV eliminates degeneracy, feedback contributes minimally; when vertical resolution is sparse (OS1-16), σ_n^2 dominates; when FOV is narrow (Avia), both are critical. The indoor paired comparison corroborates this regime separation: Mid-360 HD-LIO matches baselines (0.150m), while on Avia in the same environments all baselines diverge and HD-LIO maintains 0.524m — a qualitative gap, divergence versus bounded drift, in a regime where narrow-FOV sampling yields effective rank ≤ 3 in 40% of frames.

Failure modes. On VI03 (908m corridor, 17min), sustained planar geometry makes (S1) quantitatively marginal: median $I_d=0.18$ along the longitudinal direction (vs. 1.4 on outdoor sequences), and 18% of frames violate

(S2). Theorem 1 therefore does not apply unconditionally over this trajectory; on the 82% of frames where (S2) holds, the bound from (10) predicts a UUB radius of ≈ 0.5 m, while the remaining 18% of frames contribute an unmodeled drift increment that accumulates over the 10,268-frame trajectory and produces the observed 0.608 m ATE. Surfel, σ_n^2 , and L2 removal each degrade by $\times 1.7$ – 2.2 here, while degeneracy feedback shows only $\times 1.2$, indicating that under the weakest (S1) regime the contraction mechanism reaches its assumption-limited regime: the mechanism is operating where designed, but the assumption violation fraction sets a hard lower bound on achievable accuracy.

Conclusion. HD-LIO formalizes the drift-map positive feedback loop as a per-direction linearized recursion, derives conditions for ultimate boundedness, and instantiates a three-mechanism intervention on a hierarchical surfel map with eigenvalue-perturbation-derived per-correspondence weights. It achieves 0.203 m mean ATE on 35 baseline-comparable sequences, 22% below LIO-SAM on NTU VIRAL, and is the only method to maintain bounded error on the 8 indoor Avia sequences (70°) where all six baselines diverge. Future work: iterative motion undistortion and temporal surfel invalidation.

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